PROJECT UPDATE · AUGUST 2026

Where Does the Performance Come From?

The new scope of the bruxism paper (v8): from selling an audio–EMG classifier to auditing the whole pipeline — what changed since the Scientific Reports submission, why, and where it goes next

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Sources: main_v8.tex (26 Aug) · advisor briefing (19 Aug) · audit notes cause.md and audio.md · Scientific Reports decision (3 Jul) · venue facts checked 28 Aug 2026. No number here is new.

WHERE WE WERE

v1–v3 sold a fusion classifier for bruxism

v1 — submitted to Scientific Reports

- *“Sensor-based Bruxism Detection with Dual-Branch Wavelet CNNs and Audio–EMG Data Fusion”*
- Claimed 85.0% accuracy / 80.3% F1 on four classes (movement, clench, grind, chew) — no resting class
- Audio fusion credited with +17.4 points; “real-time”, ~15,000 parameters, 12 ms per window
- Framed as clinical screening: “bruxism detection”

Rejected 3 Jul 2026 — 2 of 4 reviewers critical

- **R2 (clinician):** tooth-contact-only definition is outdated; phenotype unspecified; discuss STAB, the biopsychosocial model and TMD standard of care
- **R4 (technical):** chewing inflates the headline; no resting class, so it is not detection; “generalises to unseen individuals” over-claims N = 5; events emulated, not natural

v2–v3 — the resubmission draft (Jul–Aug 2026)

- Added quiet rest → five classes; nested leave-one-subject-out; participant-level metrics; clinical terminology sections written for R2
- Still the same story: audio + EMG fusion, 82.1% accuracy / 73.6% macro-F1 / 0.962 AUC for the fused model
- RQ2 “how much does audio add?” and RQ3 “does the dual-branch CNN beat the baselines?” remained the spine
- Every number came from a filter chain and a microphone channel nobody had measured

The pitch rested on two things nobody had measured: the signal entering the model and the microphone channel. Both turned out to be defective. The EMG survived.

HOW WE GOT HERE

Three audits, six weeks, one surviving modality

3 Jul	27–28 Jul	3 Aug	7–10 Aug	12 Aug	15–17 Aug	19 Aug	25–26 Aug
Scientific Reports rejects v1	Prototype rebuilt as a package	Mains-harmonic defect found	Confirmatory runs, corrected chain	Microphone audit	v4–v7: mic dropped, pipeline framing	Advisor briefing	v8: general framing, 20 pp

Code audit · 27–28 Jul

- Prototype used the held-out subject for early stopping and checkpoint selection, then reported it as the test score
- Window-level K-fold on 50%-overlapping windows; focal-loss and wavelet-band bugs
- The published 85% matrix is irreproducible: its total counts 595 rest windows it does not contain
- Rebuilt: sealed outer folds, 196 tests, every number regenerated from a saved prediction ledger

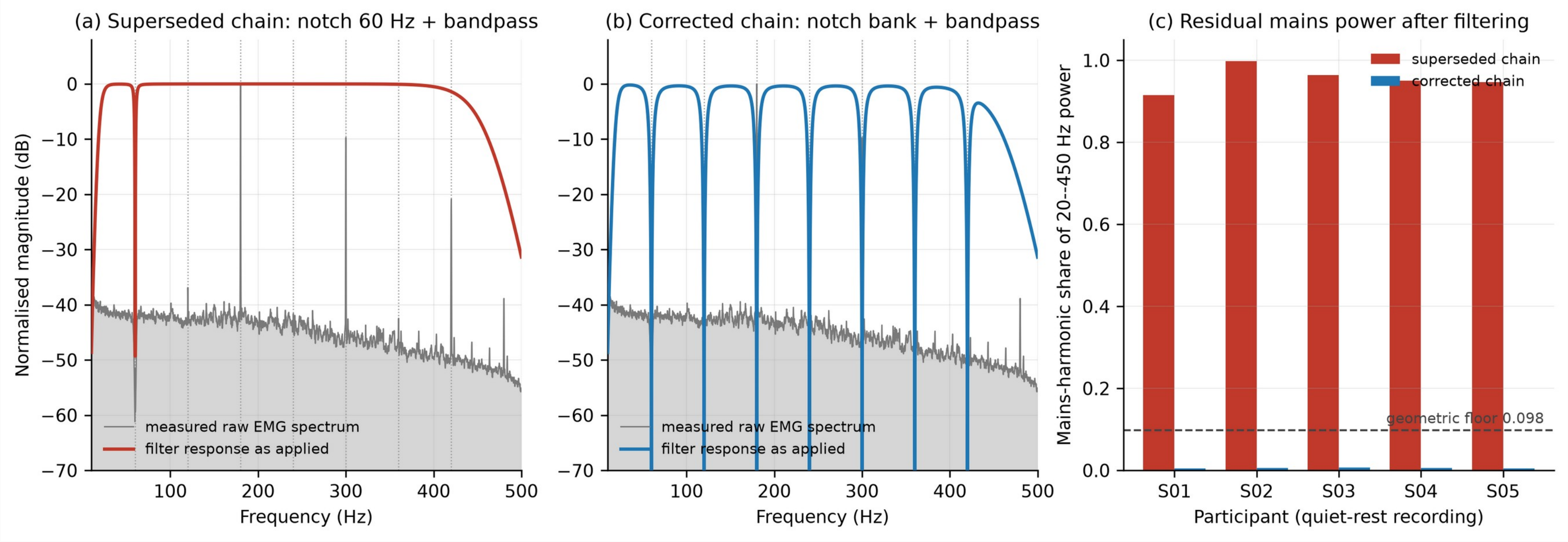
Signal audit · 3 Aug

- Acquisition hardware had already notched 60 Hz; interference sat at 180 / 300 / 420 Hz
- The textbook chain notched 60 Hz — the one frequency that was absent — and passed the rest
- 91–99.8% of in-band quiet-rest power was mains; two participants were unclassifiable
- Fix: constant-width notch bank at every harmonic, selected on an interference criterion, not on score

Channel audit · 12 Aug

- Fingerprint of every channel of every recording: EMG 100/100 distinct, microphone 37/100
- 83 recordings carry another participant’s audio, rotated — LOSO never held out the audio
- Cause: buffer reuse in the acquisition path; no second copy exists (.avi has no audio stream)
- Outcome: microphone dropped from the paper in v5 (15 Aug), put to the advisor 19 Aug; the acoustic hypothesis is untested, not refuted

The 60-Hz notch removed the one frequency that was already gone



+29.3 pts

macro-F1, 43.5% → 72.8%, same windows, folds and labels

30.2% → 6.1%

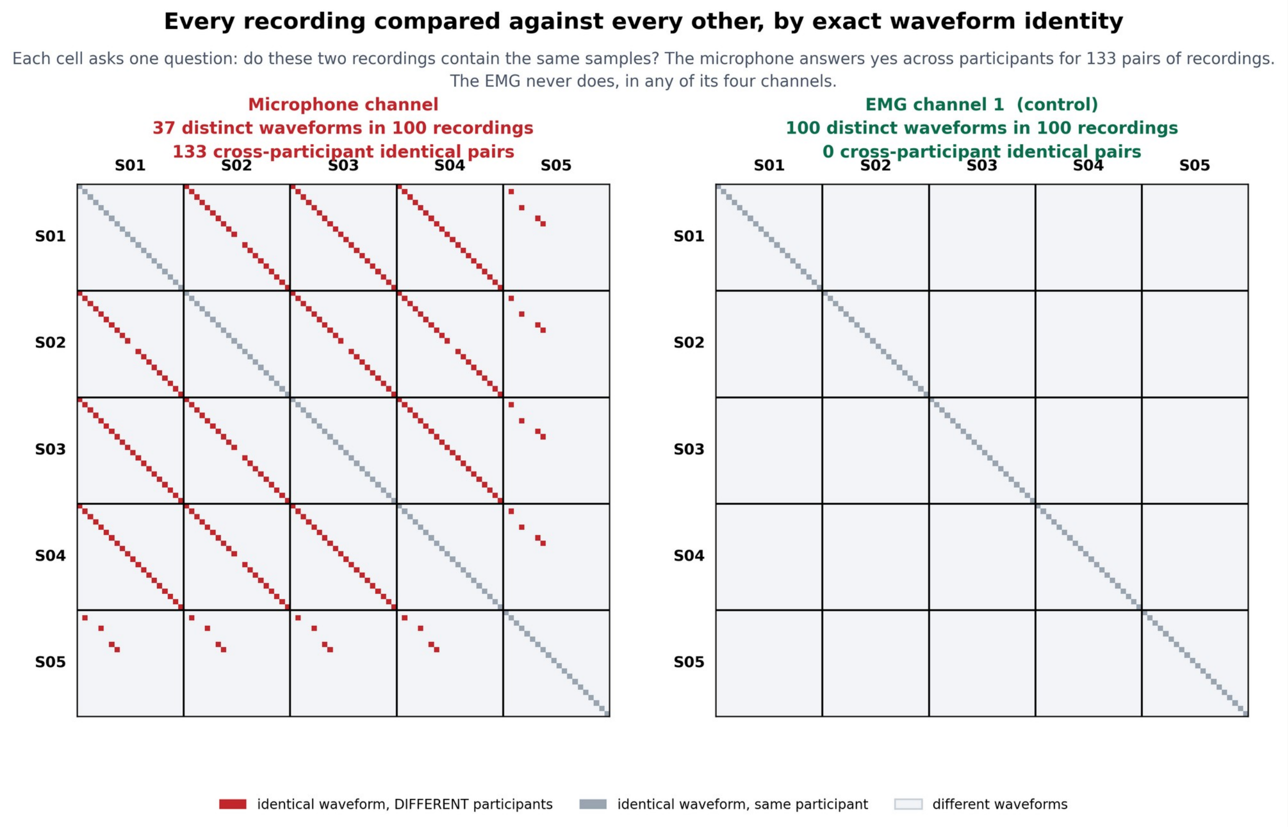
quiet-rest windows misread as clenching or grinding

2 of 5 → 0 of 5

participants at or below chance before (P1 18%, P2 5% macro-F1)

- The chain looked textbook-correct on its own response plot; the defect appeared only when the response was drawn over the measured spectrum of the recordings
- It presented as between-participant heterogeneity — the most familiar failure mode in this literature — not as a signal problem
- Three estimators of very different capacity moved 22–29 points on the same windows: a ceiling, not a model effect
- Now a pipeline stage: the residual-mains statistic is published beside every accuracy, and a band-inside-passband guard refuses to run

The microphone column is not per-participant audio



Every recording against every other, by exact waveform identity. Red: identical samples under two different participants. Left, the microphone; right, EMG channel 1 from the same files.

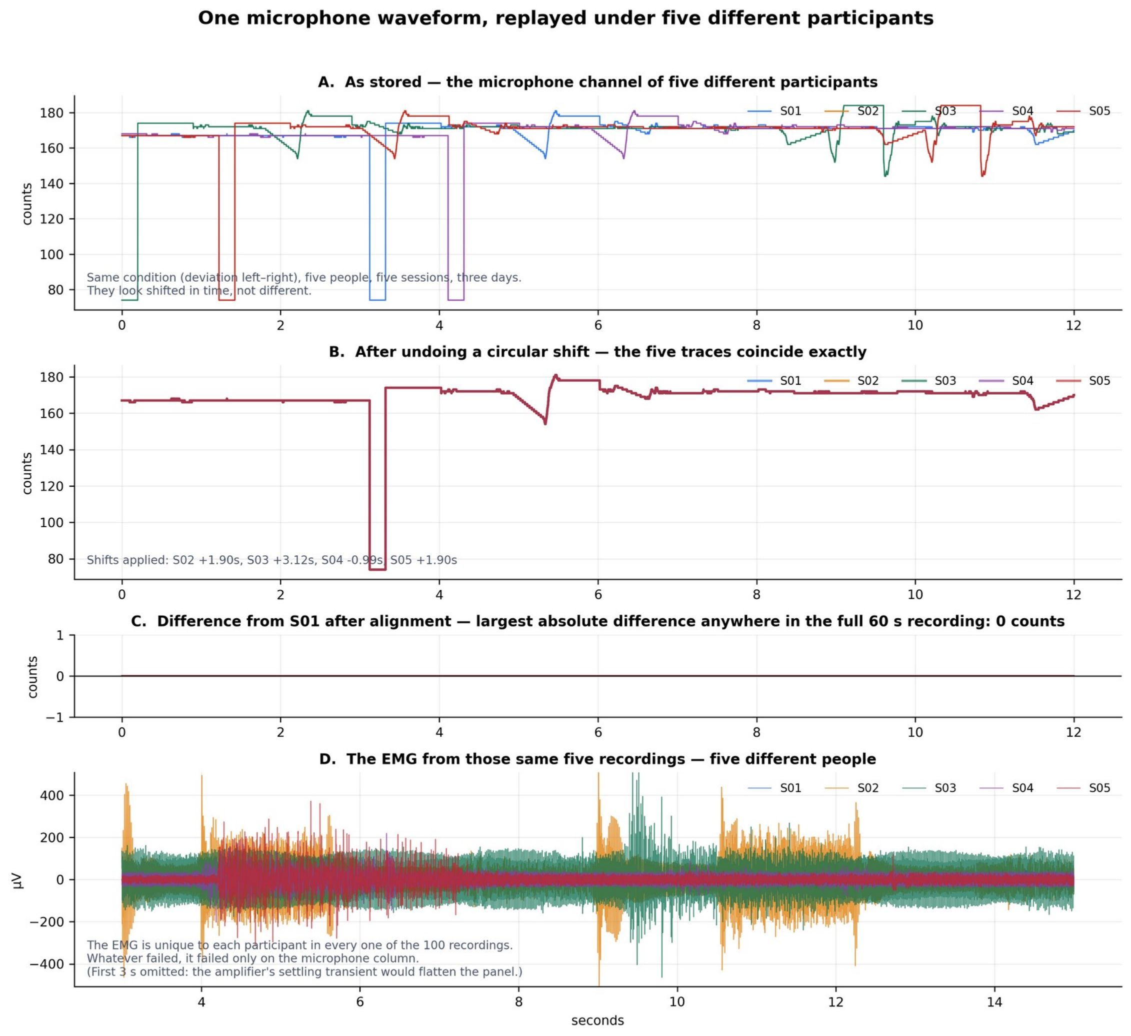
Measured from the raw CSVs (recomputed 19 Aug)

- 37 distinct waveforms in 100 recordings; 83 recordings share a waveform with a different participant; 63 of 63 tested pairs are exact circular rotations (residual 0)
- All four EMG channels: 100 of 100 distinct, zero sharing — every EMG result stands
- LOSO never held out the audio: the old fusion and audio-only numbers scored training data
- Not audio even if unique: 96% of power below 10 Hz, 1-count quantisation, clenching quieter than silent rest (0.30×)
- Not repairable: the .avi files carry no audio stream; the .npy companions have an all-zero mic column

What we did

- Dropped the microphone from every result; RQ2 recorded as untested, not refuted
- Kept the check: a rotation-invariant fingerprint of every channel on every manifest build; a run is refused if a held-out participant’s signal is in its own training set

One microphone waveform, replayed under five different participants



A · As stored. The deviation-left-right condition from five people, five sessions, three days (4–7 Aug 2025). The traces look shifted in time, not different.

B · After undoing a circular shift (S02 +1.90 s, S03 +3.12 s, S04 −0.99 s, S05 +1.90 s) the five traces coincide exactly.

C · Difference from S01 after alignment: 0 counts anywhere in the full 60-s recording. Not correlated, not similar — identical, bit for bit.

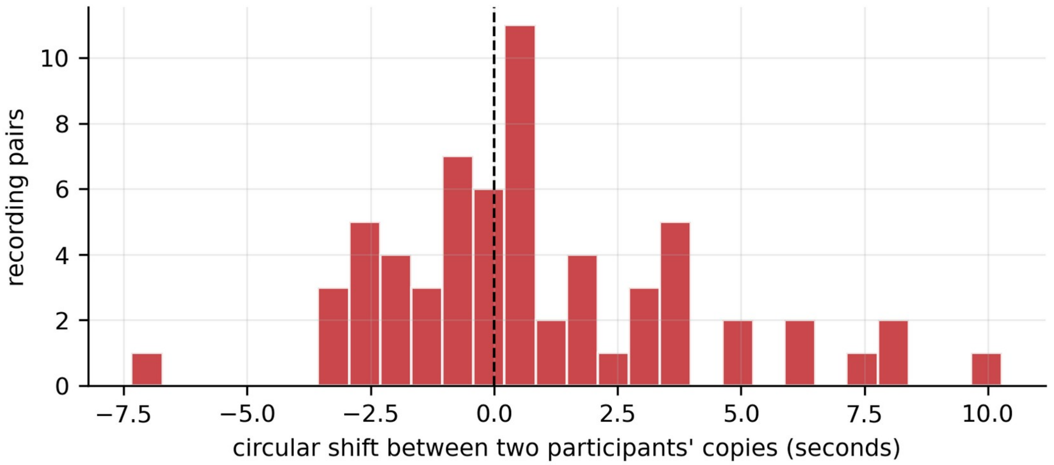
D · The EMG from the same five files is five clearly different people. Whatever failed, failed only on the microphone column.

Same +1.90 s for S02 and S05: what a direct copy of S02's column into S05's files would produce. The audit confirms it — S05's four duplicated recordings carry S02's exact sample offsets (5,586 / 69,009 / 2,279 / 69,728).

What broke: ring-buffer offsets, copies grouped by condition

How the copies are organised, and what that says about the cause

Shifts across all 63 confirmed pairs



All 63 of 63 pairs tested are EXACT circular rotations: every sample matches, bit for bit, once shifted. The shifts are small and wrap in both directions — the signature of a ring buffer whose read pointer was never reset between sessions, not of a resampling or export step.

The 20 cross-participant groups (10 largest shown)

waveform id	copies	participants	condition
4050d3ab	5	S01, S02, S03, S04, S05	deviation_left_right
ab9accf5	5	S01, S02, S03, S04, S05	bite_left
b07f4169	5	S01, S02, S03, S04, S05	incisor_clench (+1)
f51c17e4	5	S01, S02, S03, S04, S05	bite_right
3254cfeb	4	S01, S02, S03, S04	gum
6412353e	4	S01, S02, S03, S04	natural_bruxing
6994eb8a	4	S01, S02, S03, S04	cheese
6e86bdac	4	S01, S02, S03, S04	cheese
7458b218	4	S01, S02, S03, S04	protrusion_retrusion
8b1e13b0	4	S01, S02, S03, S04	carrots

Each row is one waveform stored under several participants, under the same condition. The copies are organised by condition, not by session — which is what makes this a labelled-data problem rather than merely a repeated file.

Exact rotations, small offsets

All 63 tested pairs match bit for bit once shifted. Shifts run −7.5 to +10 s and wrap both ways around zero: a ring buffer whose read pointer was never reset between sessions, not a resampling or export step.

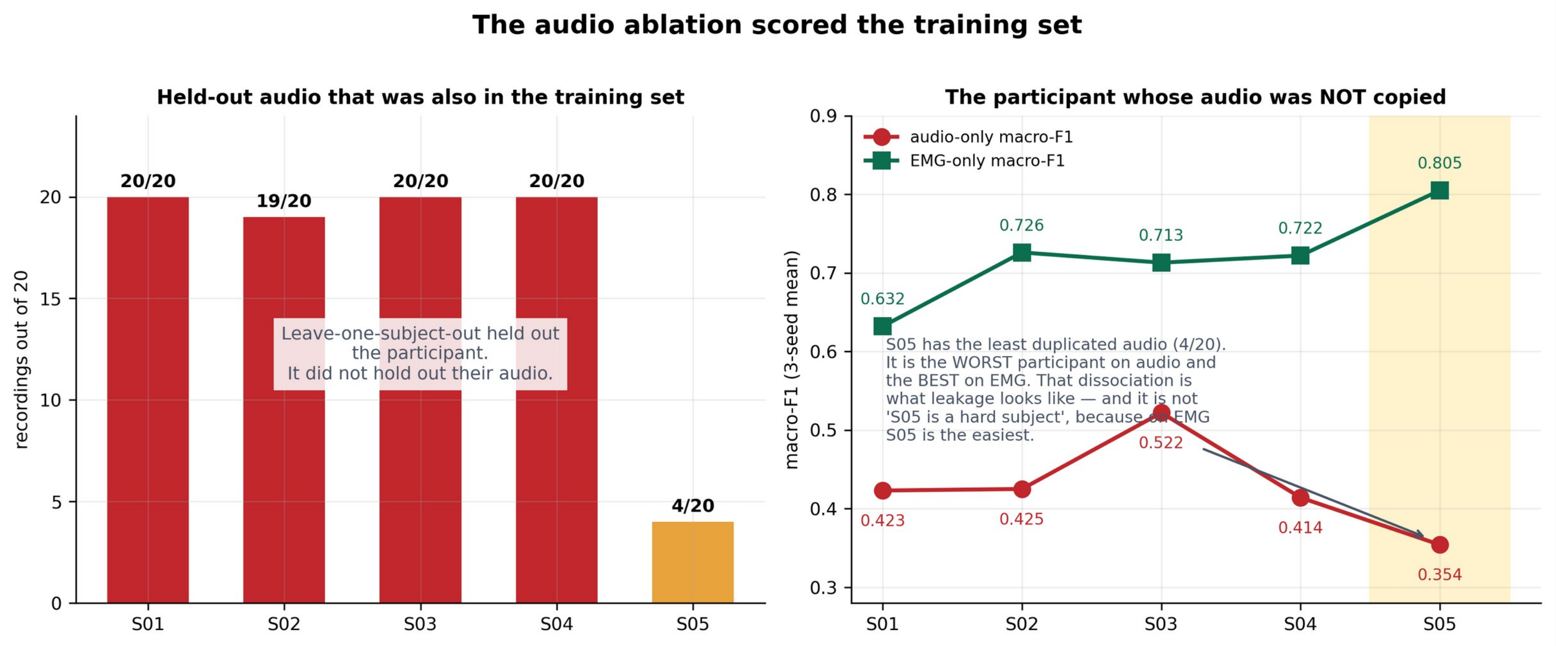
Grouped by condition, not session

Four waveforms sit under all five participants and sixteen under four, always under the same condition — which makes it a labelled-data problem, not a repeated file.

Nothing to recover from

The three .npy companions (S01) match the CSV on EMG and trigger and hold an all-zero mic column; all 100 .avi files carry zero audio streams.

The audio ablation scored the training set

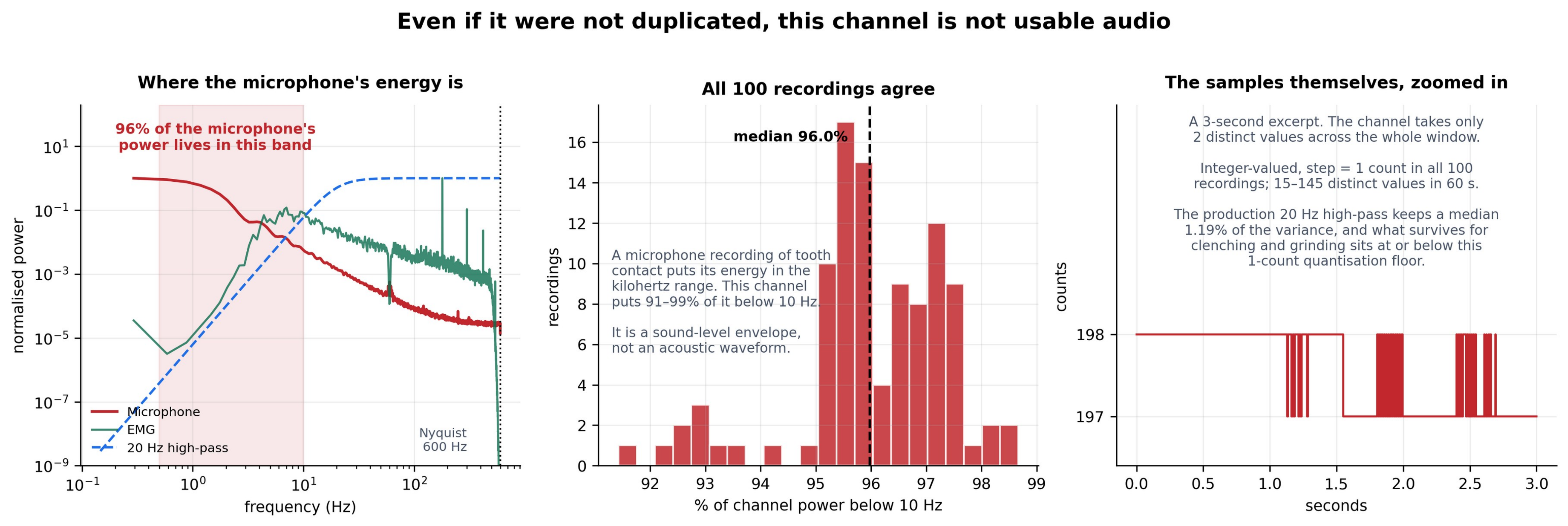


Held out	Mic dup.	Audio F1	EMG F1
S01	20 / 20	0.423	0.632
S02	19 / 20	0.425	0.726
S03	20 / 20	0.522	0.713
S04	20 / 20	0.414	0.722
S05	4 / 20	0.354	0.805

Ablation ledger modality_and_no_chewing_...cead62e4, five-class task, three-seed means (audio.md §1.6). Dup.: recordings whose mic waveform also appears under another participant.

- LOSO held out the participant, not their audio: for S01–S04 essentially every held-out audio window was also in training under the same label; the four S01–S04 rest recordings share one waveform
- S05 — the only participant whose audio was mostly not copied — is the worst on audio and the best on EMG: the dissociation leakage predicts, not a hard subject
- The one cross-condition swap behaves as predicted: S05’s protrusion recording carries the incisor-clench waveform; its 63 audio-only predictions were clench, grind or rest — 0 movement
- Withdrawn: the audio-only and fusion rows of the old Table 4, the “audio helps rest” reading, and 3–4 screening points from seven microphone features. The bias is not uniform in direction, so no corrected number can be stated

Even if it were not duplicated, this channel is not usable audio



Where the energy is

96% of the microphone's power (median; 91–99% across all 100 recordings) lies below 10 Hz. Tooth contact radiates in the kilohertz range; at 1200 Hz the Nyquist limit is 600 Hz, so that band does not exist in these files.

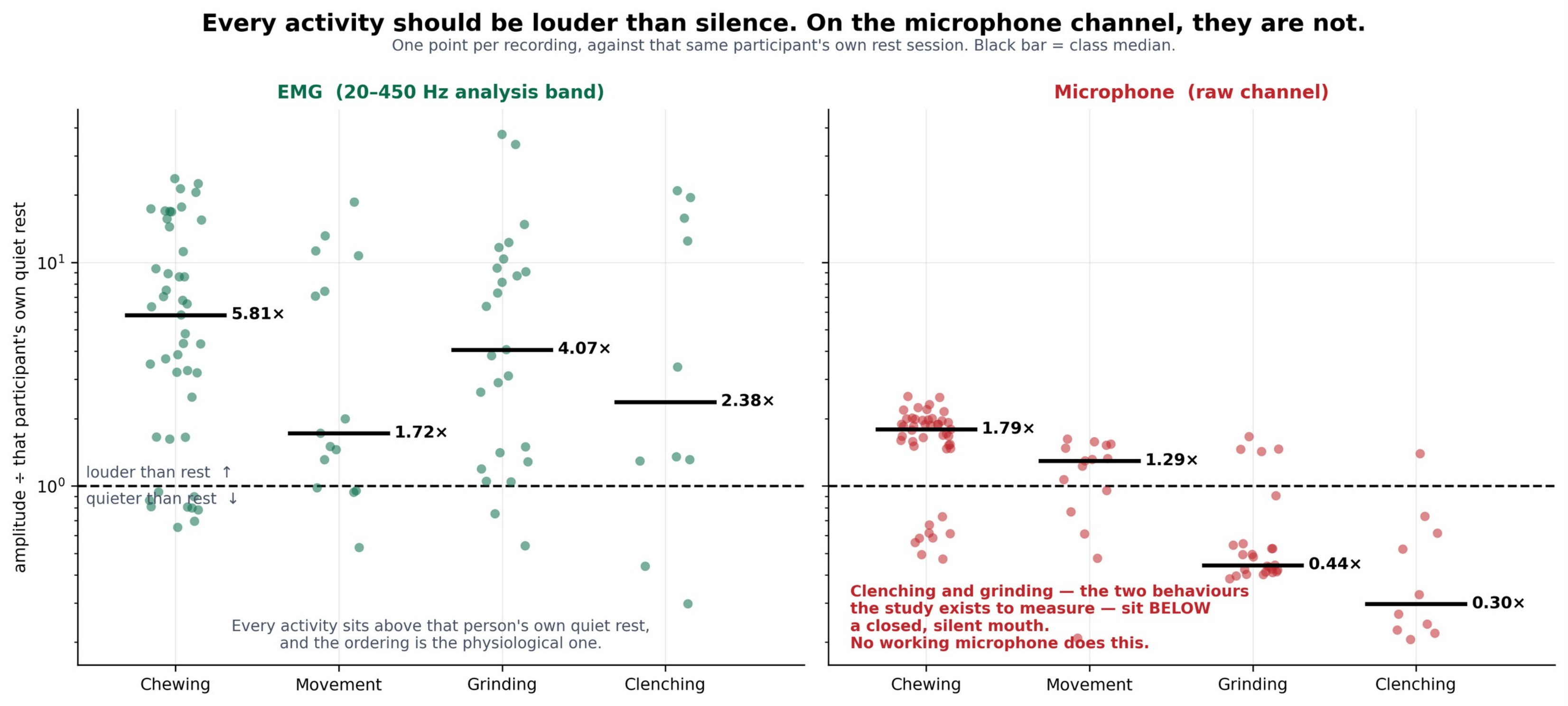
What the pipeline kept

The production 20-Hz high-pass retains a median 1.19% of the channel's variance — it removes the only band that carried condition information and keeps the band that does not.

What the samples are

Integer-valued with a 1-count step; 15–145 distinct values per minute. After the high-pass, clenching and grinding sit at or below the quantisation floor: the audio branch was reading converter dither.

Every activity should be louder than silence; on the microphone, two are not



One point per recording: each activity's amplitude divided by that same participant's own quiet-rest session. Above the line is louder than rest.

EMG (left): every activity sits above that person's rest, in the physiological order — chewing 5.8×, grinding 4.1×, clenching 2.4×, movement 1.7×.

Microphone (right): chewing 1.8×, movement 1.3× — then grinding 0.44× and clenching 0.30×. The two behaviours the study exists to measure register quieter than a closed, silent mouth. No working microphone does that.

Why it matters: this needs no signal-processing background, uses identical recordings for both channels, and is the first check to run on any new collection.

From “does our classifier work?” to “where does the performance come from?”

Where Does the Performance Come From? A Stage-by-Stage Analysis of a Biosignal Classification Pipeline, with Surface-EMG Jaw-Activity Recognition as the Case Study

Pipelines are reported through a summary accuracy and a block diagram. We vary one stage at a time on identical windows, folds and labels, and price every stage on one scale.

RQ1 How much of what the classifier learns is physiology, and how much is mains interference?

RQ2 Can four EMG channels separate quiet rest from four instructed jaw activities on an unseen participant?

RQ3 Which part of the model does the work: representation, architecture, or parameter count?

RQ4 How do those compare with choices fixed before training: window length, temporal context, normalisation scope?

KEPT

- Five-class EMG task with quiet rest · nested LOSO, 3 seeds, participant-level metrics
- Compact wavelet CNN (5,901 parameters) as the characterised reference
- The dataset and protocol — now foregrounded, with why they had to be collected

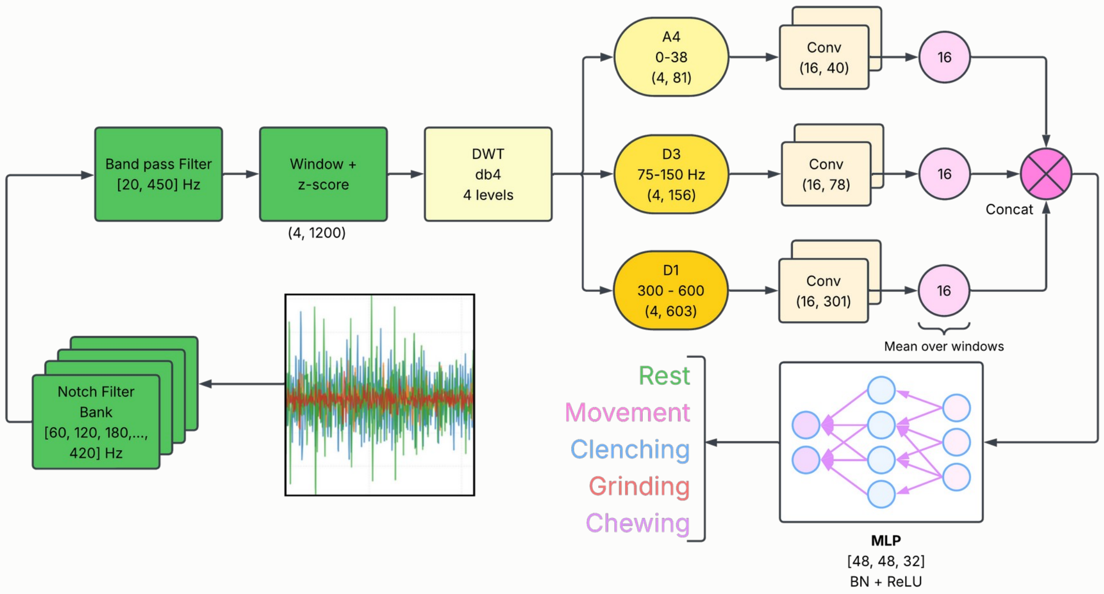
GONE

- Audio–EMG fusion and the microphone · “bruxism detection” and real-time claims
- RQ2-as-fusion and RQ3-as-“our model wins”
- The clinical / STAB / TMD sections written for Reviewer 2 (engineering venue)

NEW

- RQ1 filter contrast on identical windows · architecture sweep: representation vs. size
- RQ4 pre-model choices · the budget table (main result) · guards that refuse to run
- Positioning table (10 prior studies) · why-own-data section · a procedure that transfers to EEG / ECG / IMU

EMG alone, on participants the model never saw



Notch bank → 20–450 Hz band-pass → 1-s windows z-scored on training participants → db4 wavelet (A4, D3, D1) → three small CNN branches → MLP. 5,901 parameters, ~1 ms per window on a CPU.

81.7% accuracy

72.0% macro-F1

0.958 macro AUC

Nested LOSO · 5 held-out participants × 3 seeds · 6,173 windows · participant-level means

Class (windows)	F1	Recall	AUC
Rest (590)	83.6%	91.4%	0.958
Movement (333)	46.0%	69.8%	0.937
Clenching (799)	68.1%	66.0%	0.936
Grinding (816)	68.6%	69.5%	0.928
Chewing (3,635)	91.1%	85.8%	0.981

7.7% of quiet-rest windows land on the tooth-contact side — the closest thing to a false-alarm rate this design allows.

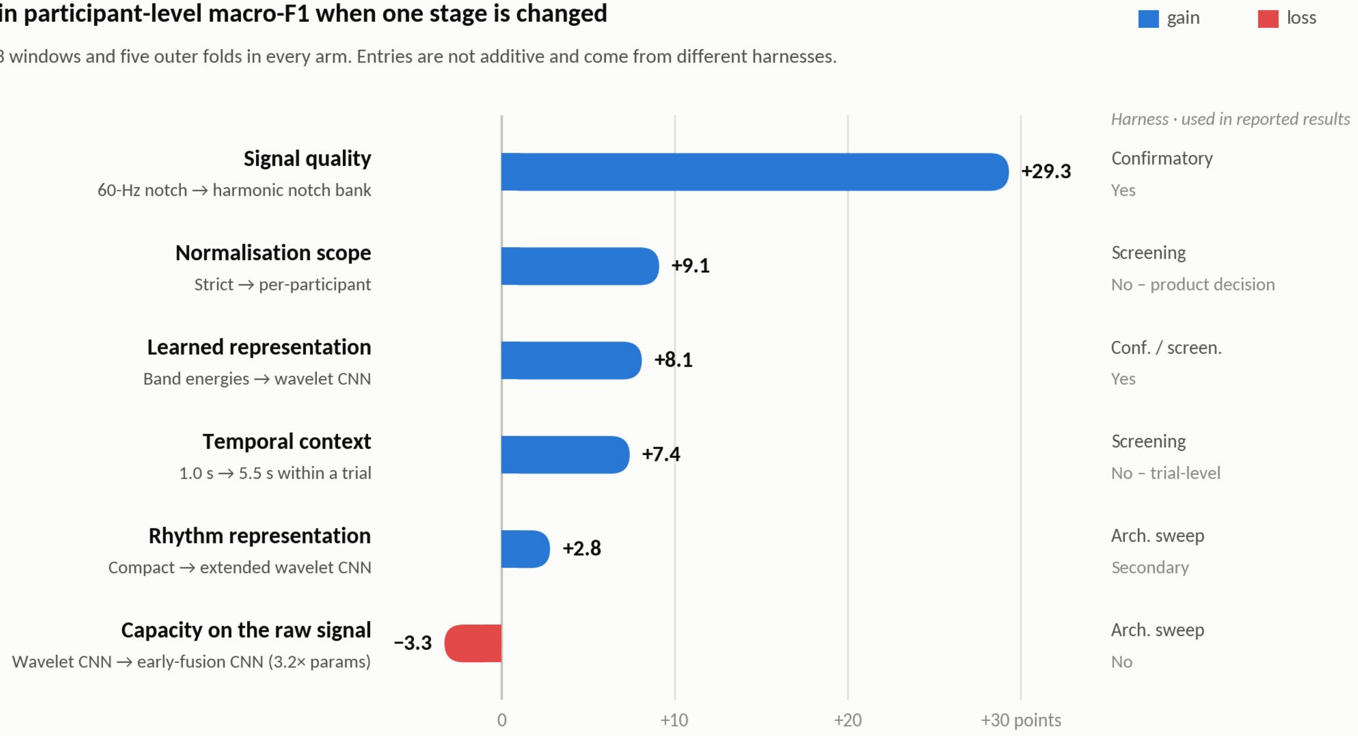
Architecture sweep — identical inputs, 3 seeds	Params	Macro-F1	AUC	Forward
Wavelet CNN, extended (time axis kept + rhythm head)	17,981	75.0 ± 1.3	0.974	2.21 ms
Wavelet CNN, compact — the reference model	5,901	72.2 ± 1.5	0.954	1.02 ms
Early-fusion CNN on the raw signals	19,029	68.9 ± 1.5	0.949	0.20 ms
Bidirectional LSTM	10,053	60.2 ± 2.6	0.898	1.16 ms

Both wavelet models above both raw-signal models; the largest model placed third. The extended model is the reference with its time axis kept — reported as secondary (2.8 points, on three of five participants).

Signal quality sets the ceiling; representation beats size

Change in participant-level macro-F1 when one stage is changed

Same 6,173 windows and five outer folds in every arm. Entries are not additive and come from different harnesses.



Below the ceiling nothing helps. Through the superseded chain the best model scored 43.5% — under the 61.3% the weakest baseline reaches on the corrected signal.

Representation beat size. The two wavelet models placed first and second; 3.2× the parameters on the raw waveform lost 3.3 points.

Two choices fixed before training — temporal context (+7.4) and normalisation scope (+9.1) — are worth as much as the architecture (+8.1) or more. Neither is a modelling decision.

Caveats, stated in the paper: entries are not additive; three rows are screening estimates, optimistic in level; the magnitudes belong to this cohort. The ordering and the procedure are what transfer.

Table 8 of main_v8.tex, redrawn. Sources: modality_and_no_chewing_...cead62e4, baselines_emg_only_20260816, outputs/screening/emg_only_*.json

WHAT WE TELL READERS

Audit your pipeline before you compare models — the five-step procedure

1

Audit the signal against what the analysis assumes

Draw every filter response over the measured spectrum of your recordings. Fingerprint every channel of every recording. Publish the interference statistic beside the accuracy.

2

Fix the data, vary one stage

Same windows, same folds, same labels in every arm — verified ledger to ledger — then change one stage at a time.

3

Price the choices made before training

Window length, temporal context, normalisation scope, label source. Here two of them were worth more than the architecture.

4

Report a budget beside the accuracy

One scale, one row per stage, each naming its harness and its baseline. Say what is not additive.

5

Keep a prediction ledger

Every number and figure regenerated from saved held-out predictions, so a figure cannot disagree with the table it depicts.

Guards that refuse rather than warn: the leakage assertion on every fold · a branch may not read a band its own filter deleted · a run may not train on a held-out participant's own waveform.

Cost: one pass over the data. Both of our defects had the same shape — a correct check scoped one step too narrowly: to the filter rather than the data, to the file rather than the dataset.

WHAT IT MEANS FOR THE GROUP

The EMG-only paper is stronger, and the audit helps the NIH direction

Why the reframe is the right call

- A LOSO number on a channel that was not held out is the kind of result that gets papers corrected; a caveat cannot fix a mislabelled number
- The EMG paper has a clear question, a quantified answer, and a method other groups can adopt in one pass over their own data
- Two independent measurement-chain defects found and fixed in one dataset — with the checks that catch them — is a contribution, not a confession
- The v8 draft is complete: 26 pp, references from p. 20, every number traced to a saved artifact

Why the audit strengthens a mic + EMG grant

- Nothing here refutes the acoustic hypothesis: these files contain no usable audio, so it is untested
- It removes a latent leakage artifact before it could sit in a Preliminary Studies section
- Concrete rigor-and-reproducibility evidence: a measurement, a versioned policy, automated tests, an ingest gate that refuses flagged data
- It turns “we will collect audio” into a protocol-plus-validation aim; the nine-requirement spec is already written (Code/docs/audio_collection_spec.md)

PROPOSAL

1

Publish the EMG-only pipeline paper now — v8 is drafted

2

Audio re-collection pilot, n ≈ 5–10, against the spec: ≥ 16 kHz WAV on its own clock, hardware sync marker, fingerprint check at ingest

3

The pilot becomes the preliminary data for the grant’s audio aim — with the failure analysis as the justification

Where to submit: journals first

Journal	JIF	Model · fee	Speed (publisher-stated)	Why it fits	Read
IEEE J. Biomedical & Health Informatics	7.7	hybrid	n.s.	Sensor-to-decision pipelines with rigorous validation; the EMBS readership	★ first choice
IEEE Trans. Biomedical Engineering	4.4	hybrid	n.s.	Strongest line on a CV; lead with the measurement-validity contribution	selective, slow
Physiological Measurement (IOP / IPEM)	2.5	hybrid	49 d to first decision (median)	Scope is “new methods of measurement and their validation” — the audit framing	good fit
IEEE Open J. Eng. in Medicine & Biology	3.1	OA · US$2,160	n.s. (“fast”)	EMBS-branded, deliberately fast; citable before the grant is written	speed play
IEEE Access	4.2	OA · US$2,160	n.s.	Broad, fast, no page limit (≤ 20 pp advised); less selective	speed play
Computers in Biology and Medicine (Elsevier)	6.3 (2024)	OA · US$3,080	n.s.	Computational-methods audience; pipelines and validation in scope	good fit
Biomedical Signal Processing and Control (Elsevier)	5.7	hybrid	n.s.	Filter / spectrum finding and the wavelet comparison squarely in scope	strong second
J. Electromyography & Kinesiology (Elsevier)	2.7	hybrid	n.s.	EMG-methods readership — where the harmonic-notch finding lands hardest	if EMG leads

Also considered (JIF 2025): Computer Methods and Programs in Biomedicine 6.4 · Medical & Biological Engineering & Computing 3.1 (an MBEC draft exists) · IEEE Sensors Journal 4.5 (8.8 wk to e-publication) · Journal of Oral Rehabilitation 3.8 (8 d to first decision; strategic — the advisor’s community; needs reframing). Journals take rolling submissions: there is no deadline to hit.

JIF = Clarivate Journal Impact Factor, JCR 2025 (released 17 Jun 2026), cross-checked against publisher pages where shown; CBM’s 2025 value was not yet listed at the check date. Fees are 2026 open-access charges (IEEE: −5% members, −20% society members). “n.s.” = not stated by the publisher.

Conference calendar: none in New England; the US options and their deadlines

Conference	Next edition	Where	Paper deadline	Proceedings	Against the three conditions
IEEE EMBS BHI 2027	Nov 2027	USA, city TBD	expected Jun 2027 (2026: 12 Jun → 3 Jul)	IEEE Xplore	watch — US confirmed, city TBD
IEEE EMBS BSN 2027	TBA (2026: 10–12 Oct, Porto)	TBA	expected Jun 2027 (2026: 15 Jun)	IEEE Xplore, 4 pp	watch — 2024 Chicago, 2025 LA
IEEE/ACM CHASE 2027	TBA (2026: 4–6 Aug, Pittsburgh)	US historically	expected Feb 2027 (2026: 16 Feb)	IEEE / ACM DL	watch
MLHC 2027	TBA (2026: 12–14 Aug, Baltimore)	US historically	expected Apr 2027 (2026: 17 Apr)	PMLR	watch
NEBEC 2027 (Northeast Bioeng. Conf.)	Apr 2027, host TBA (2026: Temple)	Northeast rotation	expected Feb 2027 (2026: 20 Feb)	abstracts in recent years	watch — fails “papers” as run
AMIA 2027 Amplify	12–15 Apr 2027	Atlanta	3 Sep 2026 — open now	archived proceedings	US ✓ · New England ✗
AMIA 2027 Annual Symposium	6–10 Nov 2027	San Diego	expected Mar 2027	archived proceedings	US ✓ · New England ✗
IEEE SPMB 2027	Dec 2027 (2026: 5 Dec)	Philadelphia (Temple)	expected Jul 2027 (2026: 1 Jul)	IEEE Xplore	US ✓ · New England ✗
IEEE EMBC 2027	11–15 Jul 2027	Singapore	expected Jan–Feb 2027	IEEE Xplore	outside the US
IEEE EMBS NER 2027	7–10 Sep 2027	Milan	expected Jun 2027 (2025: 10 Jun)	IEEE Xplore	outside the US
IEEE I2MTC 2027	17–20 May 2027	Chongqing	expected Nov 2026	IEEE Xplore	outside the US
IEEE MeMeA 2027	2027, dates TBA	Rome	24 Jan 2027	IEEE Xplore	outside the US
IEEE EMBS BHI 2026	12–15 Dec 2026	Hong Kong	closed (3 Jul 2026)	IEEE Xplore	outside the US
IEEE SENSORS 2026	25–28 Oct 2026	Rotterdam	closed	IEEE Xplore	outside the US
ML4H 2026 (with NeurIPS)	6–7 Dec 2026	Sydney	~Sep 2026 (confirm)	PMLR	outside the US

Conditions: US · New England · full papers in proceedings. **None announced for 2026–27 meets all three.** “Expected” deadlines are inferred from the previous edition and must be confirmed against the 2027 call. Also checked and outside the US or region: BSN ’26 Porto, BioCAS ’26 Incheon, ICHI ’26 Minneapolis, CHIL ’26 Seattle, MLSP ’26 Atlanta, BIBM ’26 Dallas, ICMLA ’26 Michigan, CBMS ’26 Cyprus, SenSys ’26 Saint-Malo, UbiComp ’26 Shanghai, BIOSTEC ’27 Malta, HPEC ’26 Boston (HPC), URTC ’26 Cambridge (undergraduates only).

DECISIONS

What I need from this meeting

1 Approve the v8 framing and title

A general stage-by-stage analysis with EMG jaw activity as the case study. 26 pp, references from p. 20; every number traces to a saved artifact.

2 Pick the target journal

Recommendation: IEEE JBHI (JIF 7.7). OJEMB (3.1) or IEEE Access (4.2) if being citable before the grant matters more than the venue.

3 Approve the audio re-collection pilot and its acceptance criteria

The grant's audio aim rests on it; the nine-requirement spec is written and each requirement traces to an observed failure.

4 Close the open items before submission

Hardware, gain and ADC documentation (Q3) · the IRB identifier (records carry conflicting values, Q10) · data-release scope (Q11) · co-author sign-off on the dropped microphone and clinical sections.

Nothing in the EMG results changes if the answers are "yes". Everything in the microphone story changes if the pilot is not run.